

UNIT-6

Risk Management

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Risk analysis & Management

A **risk** is a **potential (probable) problem** – which might **happen and might not**

Conceptual definition of risk

- Risk concerns future happenings
- Risk involves change in mind, opinion, actions, places, etc.
- Risk involves choice and the uncertainty that choice entails

Two characteristics of risk

Uncertainty

The risk **may or may not happen**, so there are no 100% risks (some of those may called constraints)

Loss

If the **risk** becomes a **reality** and **unwanted consequences** or **losses** occur

Risk Categorization: Approach-1

- Project risks
- Technical risks
- Business risks

Sub-categories of Business risks

- Market risk
- Strategic risk
- Sales risk
- Management risk
- Budget risk



Risk Categorization: Approach-2

- **Known risks**
 - Those **risks** that can be **uncovered after careful evaluation** of
- **Predictable risks**
 - Those **risks** that are **deduced** (draw conclusion) from **past project** experience (Ex. past turnover)
- **Unpredictable risks**
 - Those **risks** that can and do **occur**, but are **extremely difficult** to **identify** in advance



Steps for Risk Management

1. **Identify** possible **risks** and recognize what can go wrong
2. **Analyze** each **risk** to estimate the probability that it will occur and the impact (i.e., damage) that it will do if it does occur
3. **Rank** the **risks** by probability and impact. Impact may be negligible, marginal, critical, and catastrophic.
4. **Develop** a contingency **plan** to **manage** those **risks** having high probability and high impact



Risk Strategies (Reactive vs. Proactive)

- **Reactive risk strategies**
 - **"Don't worry, I will think of something".**
 - The **majority of software teams** and managers **rely** on this **approach**
 - **Nothing** is **done** about **risks** until **something** goes **wrong**
 - The **team** then **flies into action** in an attempt to **correct the problem rapidly** (fire fighting)
 - **Crisis management** is the **choice** of management **techniques**
- **Proactive risk strategies**
 - **Steps** for **risk management** are **followed**
 - Primary **objective** is to **avoid risk** and to have an **emergency plan in place** to **handle** unavoidable **risks** in a **controlled and effective manner**



Risk Identification

- Risk identification is a **systematic attempt** to **specify threats** to the project **plan**
- By **identifying** known and predictable **risks**, the project manager **takes a first step** toward,
 - avoiding them when possible
 - controlling them when necessary
- **Generic Risks**
 - **Risks** that are a potential **threat** to **every** software **project**
- **Product-specific Risks**
 - Risks that can be **identified only by clear understanding** of the technology, the people and the environment, that is **specific to the software** that is to be built



Known and Predictable Risk Categories

- **Product Size**
- **Business Impact**
- **Customer Characteristics**
- **Process Definition**
- **Development Environment**
- **Technology to be Built**
- **Staff Size and Experience**



Risk Estimation (Projection)

Risk projection (or estimation) attempts to rate each risk in two ways

- The **probability** that the **risk is real**
- The consequence (**effect**) of the **problems** associated with the risk

Risk Projection/Estimation Steps

- **Establish a scale** that reflects the **perceived likelihood (probability)** of a risk.
Ex., 1-low, 10-high
- Explain the **consequences of the risk**
- **Estimate the impact of the risk** on the project and product.
- **Note the overall accuracy of the risk projection** so that there will be **no misunderstandings**



RMMM

- **RMMM** - Mitigation, Monitoring, and Management
- An **effective strategy for dealing with risk** must consider three issues
 - Risk **mitigation** (i.e., **avoidance**)
 - Risk **monitoring**
 - Risk **management** and **contingency planning**



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